Impact of Delayed School Start Time on High School Students' Attendance, Tardiness, and Academic Achievement

CENTER FOR APPLIED RESEARCH AND EDUCATIONAL IMPROVEMENT

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INTRODUCTION

High school students often seem to stay up too late at night and then have difficulty getting out of bed the next morning. Although there might be social and environmental factors that influence adolescents' sleep behavior, recent research on the sleep-wake cycle has identified changes in specific biological processes that occur with the onset of puberty that cause adolescents not only to need more sleep but also to feel sleepy at a later time (e.g., Crowley, Acebo, & Carskadon, 2007). Because the sleep-wake cycle changes as children grow into adolescents, early high school start time has been identified as an important external factor that could restrict sleep and negatively affect academic learning (e.g., Owens, Belon, & Moss, 2010).

PURPOSE

The purpose of this study was to investigate the impact of a delay in school start time on the attendance, tardiness, and academic achievement of students enrolled in a high school where the start time was recently changed from 7:35 a.m. to 8:05 a.m., a delay of 30 minutes.

LITERATURE REVIEW

- Results of the National Sleep Foundation's 2006 poll showed that 62% of adolescents in 9th to 12th grade got an insufficient amount of sleep on school nights (less than 8 hrs.). Poll results also indicated that academic performance is related to amount of sleep, with students who got an optimal amount of sleep (9+ hrs.) reporting higher grades than students who got an insufficient amount (National Sleep Foundation, 2006).
- In a meta-analysis of research on sleep and school performance, investigators found that sleep duration, sleep quality, and sleepiness were related to school performance (Dewald, Meijer, Oort, Kerkhof, & Bögels, 2010).
- Significant pre- to post-start-time-delay increases in attendance rates were identified for 9th-, 10th-, and 11th-grade students enrolled in Minneapolis high schools where start time was changed from 7:15 a.m. to 8:40 a.m. (Wahlstrom, Davison, Choi, & Ross, 2001).
- None of the pre- to post-start-time-delay comparisons carried out on actual grade point averages of Minneapolis high school students were statistically significant (Wahlstrom, 2002).

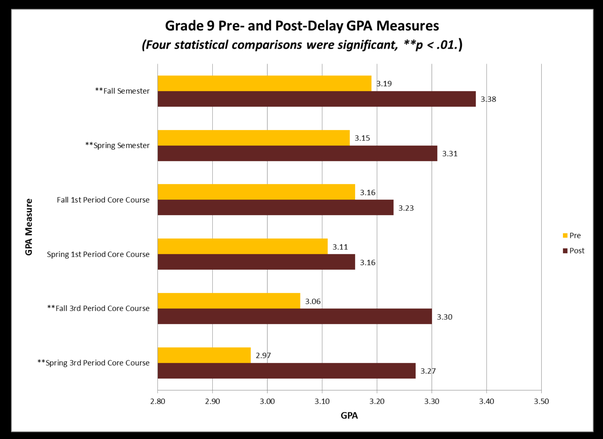
METHODS

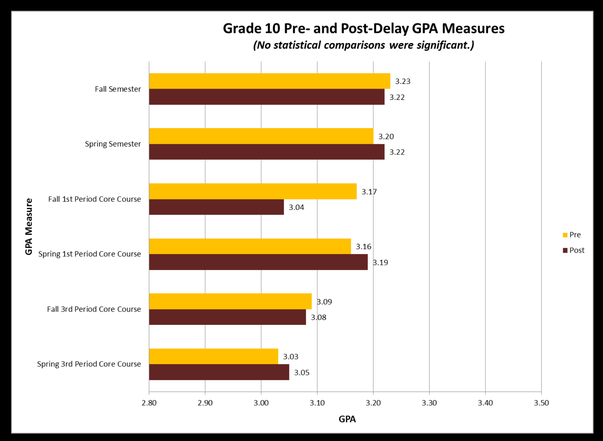
The Planning and Assessment Department of the Boulder Valley School District in Boulder, CO, provided data for Fairview High School students for the school year before (2010-11) and after (2011-12) the start time change (7:35 a.m. to 8:05 a.m.). The outcome measures investigated in this study were attendance rate, tardies to school, ACT score, and six GPA measures: Fall semester GPA, spring semester GPA, 1st and 3rd period core course GPA's for the fall semester, and 1st and 3rd period core course GPA's for the spring semester. The core course GPA was comprised of grades earned in English, math, science, and social studies. All GPA's were unweighted.

RESULTS

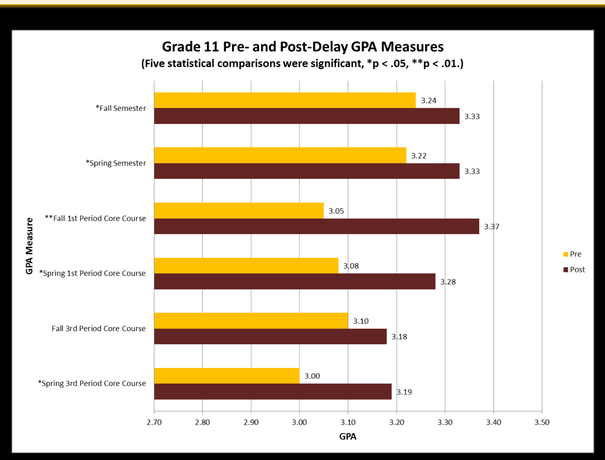
Statistical tests were applied to independent samples, and statistical significance was based on a nondirectional type I error probability of .05. Practical significance was addressed by calculating a standardized effect size, Cohen's *d*, for each result. For attendance, tardiness, and GPA measures, analyses were carried out on data for each grade level separately as well as on data aggregated across grade levels. The ACT analysis was carried out on grade 11 data only.

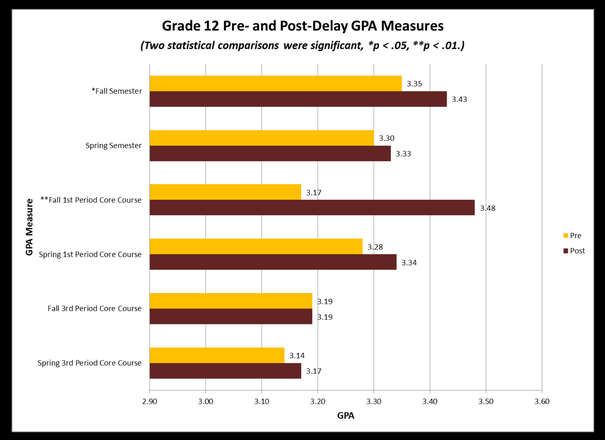
Attendance rate. None of the pre- versus post-delay comparisons were statistically significant. Mean pre and post attendance rates ranged from .93 to .97. Cohen's *d* ranged from -0.08 to 0.06.
Tardiness. Only the grade 9 comparison was statistically significant. The pre-delay mean was significantly greater than the post-delay mean (pre = 2.47, post = 1.88, $p < .05$, $d = -0.14$), indicating a decrease in grade 9 students' tardiness. Cohen's *d* for the tardiness comparisons ranged from -0.14 to -0.02.
ACT. The ACT composite difference was not statistically significant, (pre = 25.8, post = 26.0, $d = 0.03$).
GPA measures. All six across-grade comparisons on pre-post GPA measures were statistically significant, $p < .05$. Cohen's *d* ranged from 0.09 to 0.14. The graphs display results of pre-post GPA comparisons for each grade level.





RESULTS (continued)





CONCLUSIONS

The start-time delay had no impact on attendance rates of students enrolled in the high school that was investigated in this study, and the only significant pre-post difference regarding tardiness was a decrease for 9th graders. In general, however, the students exhibited very good attendance and tardiness behavior before the start time delay, leaving little room for improvement. The primary benefit associated with a delay in school start time was an increase in GPA, with the most consistent benefits received by students in grades 9 and 11. However, with respect to practical significance, all GPA effect sizes (Cohen's *d*) were very small.

LIMITATIONS

The results of this study are based on student record data for only one high school and for only one year before and one year after the delay in school start time.

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